3.3. $n_i = 1.5 \times 10^{10}$ carriers/cm^3 @ $T = 300K$

$\#e = \text{volume} \times n_i = \pi \cdot \left(\frac{15}{2}\right)^2 \cdot (0.3) \cdot (1.5 \times 10^{10}) = 7.95 \times 10^{10}$ free e^-

3.6. Boron doper → p type $N_A = 10^{17}$ cm^{-3}

$n_p = \frac{n_i^2}{N_A}$ n_p → @ 27°C → 300K $n_i = 1.5 \times 10^{10}$ cm^{-3}

@ 125°C → 398K $n_i = BT^{3/2} e^{-Eg/2kT} = 4.72 \times 10^{12}$ cm^{-3}

n_p@300K $= 2.25 \times 10^3$ cm^{-3} & $N_A = 10^{17}$ cm^{-3}

n_p @ 398K $= 2.28 \times 10^8$ cm^{-3}

3.11. $p_n(x) = \left[10^9 - \frac{x}{W}(10^9 - 1)\right] p_{no}$ $W = 50nm$, $D_p = 12 cm^2/s$

$p_{no} = \frac{n_i^2}{N_D} = 2.25 \times 10^4$ cm^{-3}

$J_p(x) = -qD_p \frac{dp_n(x)}{dx} \cong -q \cdot 12 \left(-\frac{1}{50 \times 10^{-7}} 10^9\right)(2.25 \times 10^4)$

$= \frac{1.6 \times 10^{-19} \times 12 \times 10^{16} \times 2.25 \times 10^4}{50}$

$= 8.6$ A/cm^2

3.13. $V_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right) = 25mV \times \ln\left(\frac{5 \cdot 10^{16} \times 5 \cdot 10^{16}}{(1.5 \times 10^{10})^2}\right)$

$V_0 \cong 0.751V$

$W = \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right) V_0} = \sqrt{\frac{2 \times 1.04 \times 10^{-12}}{1.6 \times 10^{-19}}\left(\frac{1}{5 \times 10^{16}} + \frac{1}{5 \times 10^{16}}\right) \times 0.751}$

$= 0.197 \times 10^{-4}$ cm $= 0.197 \mu m$

$N_A = N_D \Rightarrow x_n = x_p = W/2 = 0.098 \mu m \cong 0.1 \mu m$

$Q_J = A q \left(\frac{N_A N_D}{N_A + N_D}\right) W = 2 \times 10^{-8} cm^2 \cdot 1.6 \times 10^{-19} \left(\frac{N_A N_A}{2N_A}\right) W \cong 1.5 \times 10^{-15} C$

3.16. $W = \sqrt{\frac{2\epsilon_s}{q}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)V_0} \xrightarrow{N_A \gg N_D} W \simeq \sqrt{\frac{2\epsilon_s}{q}\frac{1}{N_D}V_0}$

$\Downarrow$

$W = 0.2\,\mu m = 0.2\times10^{-4}\,cm \Rightarrow V_0 = \frac{W^2 q N_D}{2\epsilon_s} = 0.31V$
$\& \; N_D = 10^{16}\,cm^{-3}$

$Q_j = A q \left(\frac{N_A N_D}{N_A+N_D}\right) W \xrightarrow{N_A \gg N_D} Q_j = A q N_D W = 3.2\times10^{-15}\,C$

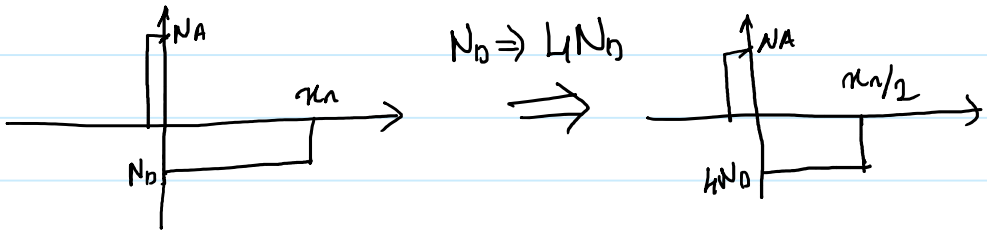
3.18 $V_{01} = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right) \Rightarrow N_A \gg N_D \gg n_i$

$\Downarrow$

$V_{02} = V_T \ln\left(\frac{N_A N_D}{n_i^2}\times 4\right) = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right) + V_T \cdot \ln 4$

$V_{02} \simeq V_T \ln\left(\frac{N_A N_D}{n_i^2}\right) = V_{01}$

$N_A \gg N_D \Rightarrow W_1 \simeq \sqrt{\frac{2\epsilon_s}{q}\frac{1}{N_D}V_{01}} = x_n, \; W_2 \simeq \sqrt{\frac{2\epsilon_s}{q}\frac{1}{4N_D}V_{01}} = \frac{1}{2}W_1 = \frac{x_n}{2}$



3.19 $V_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right) = \frac{E_{max}\cdot W}{2}$

$V_T \ln\left(\frac{N_A N_D}{n_i^2}\right) = \frac{E_{max}}{2}\sqrt{2\frac{\epsilon_s}{q}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)V_0}$

$E_{max} = \frac{2\left(\frac{kT}{q}\ln\left(\frac{N_A N_D}{n_i^2}\right)\right)^{1/2}}{\left(\frac{2\epsilon_s}{q}\left(\frac{N_A+N_D}{N_A N_D}\right)\right)^{1/2}} = \sqrt{\frac{2\,kT\ln\frac{N_A N_D}{n_i^2}}{\epsilon_s\left(\frac{1}{N_A}+\frac{1}{N_D}\right)}}$

3.25 $I = I_p + I_n \simeq A q n_i^2 \left(\frac{D_p}{L_p N_D} + \frac{D_n}{L_n N_A} \right) \left(e^{V/V_T} - 1 \right)$

$N_A \gg N_D \Rightarrow I \simeq I_p \simeq A q n_i^2 \frac{D_p}{L_p N_D} \left(e^{V/V_T} - 1 \right)$

$I_S = A q n_i^2 \frac{D_p}{L_p N_D} = 10^{-4}\,cm^2 \times 1.6 \times 10^{-19} C \times \frac{10\,cm^2/s}{10^{-3}\,cm \times 10^{16}\,cm^{-3}} \times n_i^2$

$I_S = 3.6 \times 10^{-15} A$, $V_T = kT/q \simeq 26mV$

$I = I_S \left(e^{V/V_T} - 1 \right) = 0.2m \simeq 3.6 \times 10^{-15} e^{V/V_T} \Rightarrow V \simeq 0.64V$

3.29 $C_j = \frac{\epsilon A}{W} = \epsilon_s A \Big/ \sqrt{\frac{2\epsilon_s}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) (V_0 + V_R)}$

$= A \sqrt{\frac{2\epsilon_s q}{N_A N_D} \frac{N_A N_D}{N_A + N_D}} \Big/ 2\sqrt{V_0 + V_R}$

3.30 $C_d = \left(\frac{\tau_T}{V_T} \right) \cdot I = 1pF = \frac{\tau_T}{V_T} 0.1m$

$C_d @ 0.1mA = \left(\frac{\tau_T}{V_T} \right) \cdot 0.1m = 250fF$

$\tau_T = \frac{250f}{0.1m} \cdot V_T \simeq 62ps$

$V_T \uparrow$ 25mV@ room temp.